When does an LLM trust a specialist model? A cost-aware trust-routing audit — and why "calibrated reliability interfaces" are not a free win

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Abstract

Frontier AI-for-science systems increasingly place a general LLM *above* specialist scientific models (protein, genome, single-cell), where the LLM must decide — under a verification cost — whether to **trust** a specialist's output, **verify** it, fall back to a **cheap baseline**, or **defer**. We audit that trust-routing decision and ask whether converting a specialist's reliability evidence into an explicit *calibrated reliability interface* improves routing over simply showing the raw confidence.

Across three substrates (a GEARS/Norman perturbation dry-run, a single-cell foundation-model cue, and protein-structure prediction with Boltz-2/pLDDT) the robust finding is **cautionary, not celebratory**. (1) The LLM reasoning layer is strongly **cue-sensitive**: shown any reliability signal it routes far better than with none, but it also follows misleading signals. (2) On a *validated, calibrated* signal (pLDDT, which predicts true lDDT at Pearson 0.89 for monomers), an LLM (Claude Sonnet 4.6) converts it into much better cost-aware routing — **but the benefit comes from the calibrated *information*, not a directive recommendation**, and **packaging it as a "reliability card" does not robustly beat showing the raw confidence number** (Δnet +0.025, 95% CI [-0.037, +0.087]). (3) The card is **model-dependent and can backfire**: Claude Opus 4.8, a more risk-averse model, **over-verifies** under the card (97.5% verify rate) and is *significantly worse* than with raw confidence (Δnet -0.225, CI [-0.325, -0.113]), replicating the "stronger reasoning $\neq$ better cost-aware orchestration" pattern. We conclude that **raw calibrated confidence is the robust lever; prompt-visible reliability *interfaces* are not a free win and are model-dependent** — presentation alone is insufficient, motivating enforcement-based (tool/MCP/post-training) routing.

1. Question

> What does an LLM reasoning layer trust when it reads a specialist scientific model's output, and can that trust be made calibrated, auditable, and useful under verification cost?

Action set per item: `trust_sfm | verify_assay | default_baseline | defer`. Reward: `net = correct - lambda*assays` (λ = verification price; primary $\lambda = 0.5$).

2. What each phase found

- **Phase 0 (GEARS/Norman dry run, 642 non-leakage Sonnet requests).** Routing is strongly cue-sensitive and only weakly calibrated; additive-disagreement evidence helps, misleading "reliability cards" hurt, and self-reported rationales frequently cite unavailable evidence.
- **Phase 1 (single-cell scFoundation cue).** NO-GO. The scFoundation-derived edge cue predicts GEARS wrongness at AUROC 0.599 vs 0.576 for a random same-gene control — near-noise. A specialist-metric check found GEARS ≈ a cheap additive baseline (replicating Ahlmann-Eltze et al. 2025). You cannot study calibrated trust where the signal carries no validated information.
- **Phase 2 (protein structure, Boltz-2 / pLDDT).** A substrate where the specialist is excellent *and* emits a validated calibrated confidence — the focus of this report.

3. Phase 2 methods (brief)

- **Substrate.** 80 RCSB targets released 2026-06-17 (after Boltz-2's 2023-06 training cutoff → leakage-safe), 40 monomer + 40 complex; sequences + reference structures from RCSB; structures predicted with Boltz-2 on an A100/A40 (Cayuga).
- **Truth.** Superposition-free CA-lDDT of prediction vs experimental reference (Mariani et al. 2013), monomer single-chain and complex all-chain (interface-inclusive). *Caveat: home-grown CA-lDDT, not yet validated against OpenStructure; complexes use lDDT, not DockQ.*
- **Calibrated signal.** pLDDT→P(wrong) via leave-one-out isotonic regression. pLDDT predicts lDDT at **Pearson 0.89 (monomers), 0.16 (complexes)** — the designed monomer→complex calibration gap, confirmed.
- **Offline gate.** A deterministic "verify iff calibrated-risk > λ" policy must beat trust-all + shuffled/inverted controls before any LLM call. *Caveat: it is degenerate at the standard lDDT ≥ 0.7 cutoff (Boltz-2 is correct on ~95% of recent targets) and was made to pass at lDDT ≥ 0.9 — a post-hoc choice; signal AUROC there is 0.89.*
- **LLM pilot.** A balanced **40-target subset (20 monomer + 20 complex)** of the 80-target benchmark, stratified across the lDDT range, × 5 arms = 200 requests, Sonnet 4.6 and Opus 4.8, scored against held-out lDDT-truth. (The full 80 targets are used for the offline calibration gate.) Arms: `no_signal`, `raw_plddt_shown`, `calibrated_risk_shown_no_recommendation` (information without a directive), `calibrated_interface_s` (risk + recommended action), `inverted_reliability_interface_control`.

4. Results

4.1 Sonnet ($\lambda = 0.5$, lDDT ≥ 0.9; target-bootstrap 95% CIs, seed 13)

arm	net/target	Δ vs no_signal (95% CI)
calibrated_interface_shown	0.738	+0.487 [+0.338, +0.625]
calibrated_risk_shown_no_recommendation	0.725	+0.475 [+0.325, +0.613]
raw_plddt_shown	0.713	+0.463 [+0.300, +0.613]
inverted_reliability_interface_control	0.388	+0.138 [+0.025, +0.250]
no_signal	0.250	—

- **Any confidence cue robustly beats no-signal** (+0.46–0.49, CIs exclude 0). With no signal Claude defers/over-verifies ("can't assess reliability").

- **A4 — the benefit is informational, not directive.** calibrated_interface vs calibrated_risk_*no_recommendation*: Δ +0.013, CI [+0.000, +0.037] $\approx$ 0.
- **The calibrated card does NOT robustly beat raw pLDDT.** calibrated_interface vs raw_plddt: Δ +0.025, CI [−0.037, +0.087] — crosses zero.
- **Regime-appropriate caution.** Under the card, verify rate is 0.55 on complexes vs 0.20 on monomers — more checking exactly where pLDDT calibration degrades.
- **Cue-sensitivity persists.** The inverted card degrades routing 0.738 $\rightarrow$ 0.388.

4.2 Cross-model (Opus 4.8, same 200 requests)

arm	Sonnet net	Opus net	Sonnet verify	Opus verify
raw_plddt_shown	0.713	0.713	0.225	0.275
calibrated_interface_shown	0.738	**0.487**	0.375	**0.975**
no_signal	0.250	0.150	0.500	0.300

Opus calibrated_interface vs raw_plddt: **Δ −0.225, CI [−0.325, −0.113] — robustly worse**. The risk-framed card triggers blanket over-verification in the more risk-averse model (it verifies even confident monomers). A4 holds for Opus too (card vs no-recommendation = 0).

4.3 Robustness

The Sonnet/Opus conclusions are **stable across lDDT cutoffs 0.5–0.9** (`results/phase2_preflight/cutoff_ro` calibrated_interface − raw is ≈ 0 for Sonnet and ≈ -0.25 for Opus at every cutoff — so the post-hoc 0.9 choice affects only the offline-gate variance, not the pilot conclusions. Across λ = 0.2/0.5/0.8 the lower arms keep their order; the only change is that calibrated_interface and raw_plddt swap at the top under high cost (card−raw: +0.070 / +0.025 / −0.020 at λ = 0.2/0.5/0.8) — which further supports that the card does not robustly beat raw confidence.

5. Interpretation

The robust, cross-phase, cross-model claim:

> The LLM reasoning layer is **cue-sensitive**; surfacing **raw calibrated confidence** is the robust lever for better cost-aware routing; repackaging it as a calibrated **reliability *interface*** is **not a free win and is model-dependent** — neutral for Sonnet and harmful for the more risk-averse Opus, which over-verifies. The benefit, where present, is **informational, not directive**. Presentation alone is insufficient; **enforcement** (tools, MCP constraints, or post-training) is the natural next lever.

This extends Turpin et al. (2023) — LLMs follow cues regardless of validity — into a constructive, *calibrated-signal* regime: a correctly calibrated cue helps (via its information), a misleading one hurts, and a "reliability card" framing can over-trigger a risk-averse model.

6. Relation to prior work

- **Proto / EvoDesign (Merchant, Guo, Viggiano, ... Hie; bioRxiv 2026.06.22.733870) — the most directly relevant system.** Proto is a high-level programming language for generative biology: an LLM/agent composes ~120 specialist models (Evo2, ESM3, AlphaFold3, Boltz-2, AlphaGenome, ProteinMPNN, …) into multi-objective design programs, treating each specialist's confidence as an energy to minimize (pi(x) ~ p(x)*exp(-f(x)/T)). It is the strongest existing validation of the hybrid "reasoning layer above specialist models" stack this project assumes – but it trusts specialist confidence UNCONDITIONALLY: no calibrated trust/verify/defer decision, no verification price, no audit of the reasoning layer's trust. Proto itself reports the gap we study – "structure-prediction metrics are plausibility filters, not guarantees of function"; "no single in-silico metric separated functional from non-functional designs" – and routes around it with multi-objective consensus plus tens of wet-lab tests. Two of its constructions are concrete,* uncalibrated* precedents for our actions: (i) its 8-stage CRISPR filter cascade runs cheap filters first and expensive AlphaFold3 last with short-circuiting – a hand-built, cost-ordered verify-vs-defer pipeline (our net = correct - lambda*assays objective, minus the calibration); (ii) its multi-oracle structure consensus (require Boltz-2 AND AF2 AND AF3 to agree) is an uncalibrated verify-by-consensus. Our contribution is the calibrated, cost-aware, pre-registered trust layer that Proto-class agentic systems currently lack.

- **Cue-following / unfaithfulness:** Turpin et al. 2023. We add a cost-aware routing game and a *validated-calibrated* cue.
- **Cost-aware routing / selective prediction:** ASPEST (Kim et al. 2023), Adaptive-RAG, FrugalGPT/RouteLLM, learning-to-defer (Madras 2018; Mozannar & Sontag 2020). We route over a *fallible scientific specialist* under an assay cost.
- **Specialist soundness:** Ahlmann-Eltze, Huber & Anders 2025 (perturbation FMs ≈ linear baselines — why single-cell failed here); Jumper et al. 2021 and the McGuffin-group calibration audit 2024 (pLDDT calibration — why protein structure worked); Cheng et al. 2023 (AlphaMissense — a runner-up substrate).

7. Limitations

(a) Offline-gate PASS engineered post-hoc (lDDT $\geq$ 0.9 + in-distribution LOO calibration); (b) home-grown CA-lDDT, unvalidated vs OpenStructure, complexes not DockQ; (c) low-stakes substrate (~95% correct at lDDT $\geq$ 0.7); (d) the model-visible packet contains only confidence (+ regime), so "uses the signal" is partly forced; (e) n = 40, single primary λ, single MSA source, one run per model, default-False template baseline. A pre-registered confirmatory redo addressing all of these is specified in `PHASE2_PREREGISTRATION.md`.

7b. Post-publication validation (2026-06-23, on-cluster)

Three follow-up checks on Cayuga directly test the limitations above against gold-standard tools (artifacts under `results/phase2_{ost,dockq,leakage}_*`):

- **Monomer truth — confirmed.** OpenStructure all-atom lDDT vs the home-grown CA-lDDT on the 40 monomers: Pearson **0.99**; pLDDT→OST-lDDT = **0.894**, confirming the headline 0.89 monomer calibration against the gold standard.
- **Complex truth — corrected.** The all-chain CA-lDDT is intra-chain-dominated, so the reported pLDDT→complex "calibration collapse" (0.16) was largely a **metric artifact**. Against gold-standard **DockQ** (40 complexes): pLDDT→DockQ = **0.44**, and the proper interface confidence **ipTM→DockQ = 0.77** — the specialist *is* well-calibrated on complexes; route on ipTM, not pLDDT. Re-scoring the v1 episodes with DockQ truth leaves the headline intact (interface−raw ≈ 0 for Sonnet, ≈ −0.25 for Opus).

- **Leakage — the "leakage-safe" claim does not hold.** An MMseqs2 search of all 80 targets vs the full PDB, date-filtered via RCSB, finds that ~**70% (56/80 at a 2024 cutoff) have a \$ \$90%-identical pre-cutoff homolog**. Release-date curation did not prevent training leakage; the ~95% base rate is partly memorization. The pre-registered redo's homolog dedup is therefore empirically required, and the substrate must be re-curated for genuinely low-homology folds.

Net: the routing **headline survives**, the monomer metric holds, the complex-calibration story is corrected, and the substrate's leakage is the principal reason to execute the confirmatory redo rather than lean on v1.

Redo update (executed). A genuinely leakage-controlled set was then curated (MMseqs2 vs full PDB + RCSB dates: **92% of recent post-cutoff depositions carry a \$ \$30% pre-cutoff homolog**; 21 non-redundant low-homology targets survive) and predicted with Boltz-2. It is **100%/100% correct at the pre-registered cutoffs** (monomer lDDT ≥ 0.7, complex DockQ ≥ 0.23). So Boltz-2 succeeds on recent novel folds too: the **low-stakes problem is intrinsic to recent high-resolution structures, not an artifact of leakage** (the redo separates the two confounds). By the project's own NO-GO discipline a full *confirmatory* pilot cannot run on a 100%-base-rate substrate — a real routing substrate needs specialist *failure* cases (harder modalities), motivating the enforcement-layer pivot. A bounded 3-model demo (Sonnet 4.6, Opus 4.8, GPT-4.1) makes the consequence concrete and is identical across all three: `calibrated_interface $-$ raw_plddt` net = -0.476 on the saturated substrate (the risk card triggers blanket over-verification where trust-all is free) vs **+0.05** only once stakes are manufactured (lDDT ≥ 0.9) — fresh, leakage-controlled, cross-model evidence that **reinforces the headline** (raw confidence is the robust lever; the reliability interface is not a free win). See `results/phase2_redo_curation/`.

8. Reproducibility

All code, tests (163 passing), and compact result artifacts are in the repo under `experiments/trust_cue_attribut`
Key artifacts: `results/phase2_preflight/{boltz_contract,boltz_smoke_report,calibration_gate,`
`calibrated_gate,interface_pilot_score,interface_pilot_robustness, interface_pilot_score_opus,i`
Pipeline: `phase2_curate_pdb.py` → Boltz predict → `phase2_truth.py` (CA-lDDT) → `phase2_records.py` → `phase2_calibrated_gate.py` → `phase2_interface_pilot.py` → `phase2_score_episodes.py` / `phase2_robustness.py`. Large JSONL (requests, episodes, structures) stay ignored under `hpc_outputs/`.

How to cite

See `CITATION.cff`. This report is intended for archival via Zenodo (GitHub release → DOI); the pre-registration is intended for OSF.